CHANGE OF BASIS VECTORS

Given two bases for R^2:

$$\mathcal{B} = \{(1,0),(0,1)\}$$

$$\mathcal{B}' = \{(1,2),(3,3)\}$$

Construct the Change of Basis Matrix

Express the new basis vectors in terms of the old basis:

$$(1,2) = 1(1,0) + 2(0,1)$$

$$(3,3) = 3(1,0) + 3(0,1)$$

The change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$ is:

$$P_{\mathcal{B}\to\mathcal{B}'} = \begin{pmatrix} 1 & 3 \\ 2 & 3 \end{pmatrix}$$

Find the Inverse

Calculate the inverse of $P_{\mathcal{B}\to\mathcal{B}'}$ to get $P_{\mathcal{B}'\to\mathcal{B}}$:

$$P_{\mathcal{B}'\to\mathcal{B}} = (P_{\mathcal{B}\to\mathcal{B}'})^{-1} = \begin{pmatrix} -1 & 1 \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$

Transform the Coordinates

Given a vector z with coordinates $[z]_{\mathcal{B}'} = (9,8)$ with respect to $\mathcal{B}'$, the coordinates of z with respect to $\mathcal{B}$ are:

$$[z]_{\mathcal{B}} = P_{\mathcal{B}'\to\mathcal{B}}[z]_{\mathcal{B}'} = \begin{pmatrix} -1 & 1 \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 9 \\ 8 \end{pmatrix} = \begin{pmatrix} -1 \\ \frac{10}{3} \end{pmatrix}$$